

Section 1: Metadata

to be filled by the student

1.1. Project Information

TypeCraft: An Offline Typing Tutor Game for Developing Keyboard Literacy in The Bridge School Students	
Start: 18 th May 2026	End: 18 th August 2026

1.2. Student(s) Information

Name:	Rayyan Aamir Alvi	Shaheer Qureshi
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Batch:	Computer Science 2028	Computer Science 2028 CE 2028

1.3. Client Information

Organization:	The Bridge School
Contact:	+92 334 3133500
Branch 01:	KE 481/A, Mughal Hazara Goth, Block 4A, Near Kamran Chowrangi, Gulistan-e-Jauhar, Karachi
Branch 02:	DNK Campus, House A-36, Block 11, Gulistan-e-Jauhar, Bhango Gabol Ghot, Near Kamran Chowrangi, Karachi
Supervisors:	1) Miss Heera Abid +92 315 2057707 2) Miss Farah Mehdavi +92 324 1266046
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Section 2: The Project

to be filled by the external supervisor

2.1. Project Description

Please provide a brief introduction of the project including its scope.

This project aims to design and develop an interactive offline typing tutor application specifically for students studying in an NGO-supported school environment. The purpose of the system is to improve students' keyboard literacy, typing speed, typing accuracy, and overall confidence in using computers through a structured, educational, and child-friendly learning platform.

In many underprivileged educational environments, students have limited access to proper computer literacy training and often lack exposure to structured typing practice. As a result, students commonly develop inefficient typing habits such as incorrect finger placement, overreliance on a few fingers, poor typing posture, and slow typing speed. Existing typing platforms are often internet-dependent, resource-intensive, difficult for younger students to use, or not optimised for older computer systems commonly available in NGO schools.

This project proposes the development of a lightweight offline typing tutor game optimised for low-end Windows 10 and Windows 11 computers running on 4th-generation Intel processors with integrated graphics and limited RAM. The application will operate entirely without internet connectivity and will provide structured typing lessons beginning with home-row keys, progressing gradually toward full keyboard familiarity through gamified, child-friendly learning activities.

The application will include a visual on-screen keyboard that highlights the correct keys and indicates which finger should be used for each key press. Real-time feedback will be provided during typing exercises, where correctly typed characters are highlighted in green while mistakes are clearly indicated in red. Performance metrics displayed to the student will include Words Per Minute (WPM), typing accuracy percentage, mistake count, completion time, combo streaks, and progress consistency across sessions.

To encourage motivation and long-term engagement, the application will incorporate gamification elements including an experience points (XP) system, progressive level unlocks, achievement badges, positive encouragement messages tailored for children, and an offline classroom leaderboard. Each student will maintain an individual profile storing their lesson history, typing statistics, badges, and progression data locally without any cloud dependency.

Complete Technology Stack

Layer	Technology	Role & Justification
Application Language	Python 3.10+	Core language for all game logic, lesson progression, XP calculation, badge evaluation, and scene management. Chosen for readability, strong Windows support, and clean PyInstaller bundling.

Layer	Technology	Role & Justification
Frontend — Rendering & UI	Pygame 2.x	Handles the entire visual layer: window creation, frame-by-frame rendering at 30 FPS, keyboard event capture, mouse input, and all on-screen animations. Every UI element — menus, lesson text, virtual keyboard, buttons — is drawn as Pygame surfaces. No HTML, no browser, and no web framework is involved, keeping the footprint minimal on low-end integrated graphics.
Frontend — UI Components	Custom Python Widgets	Reusable Button, TextInput, and virtual Keyboard components written from scratch in Python using Pygame primitives. Child-friendly sizing, large fonts, and bright color palette defined centrally in a constants file so the entire visual theme can be adjusted in one place.
Frontend — Typography	TrueType Fonts via pygame.font	Child-friendly .ttf font files loaded natively. Font surfaces are pre-rendered and cached to avoid per-frame re-rendering, a key optimisation for low-RAM systems.
Backend — Database	SQLite 3 (Python stdlib — sqlite3)	Serverless, file-based relational database storing student profiles, lesson scores, badge records, daily streaks, and leaderboard data in a single typecraft.db file. No database server installation required on school computers. Data survives power interruptions and can be backed up by copying one file to a USB drive.
Backend — Configuration & Lesson Content	JSON (Python stdlib — json)	Application settings (volume, resolution, teacher PIN hash) and all lesson content are stored in human-readable .json flat files. Teachers can open lessons.json in Notepad and add or modify lesson text without touching any code.
Backend — Security	hashlib (Python stdlib)	SHA-256 hashing is used to protect the teacher dashboard PIN. The raw PIN is never stored — only its hash — providing basic access control appropriate for a classroom environment with no external dependencies.
Backend — Audio	pygame.mixer (built into Pygame)	Lightweight audio subsystem for keystroke click sounds, error feedback tones, success chimes, and level-up fanfares. Audio files kept under 50 KB each in OGG/WAV format. A mute toggle and volume slider are accessible from the settings screen.
Packaging & Deployment	PyInstaller 6.x	Bundles the entire application — Python runtime, Pygame, all assets, fonts, sounds, and lesson data — into a standalone Windows .exe distributed as a single folder. No Python installation is required on school computers. Deployment is as simple as copying the folder via USB drive and double-clicking TypeCraft.exe.

Layer	Technology	Role & Justification
Version Control	Git (local)	Local Git repository for change tracking and rollback during development. No remote hosting required; the repository stays on the developer's machine.

The project scope also includes real classroom testing and gathering teacher feedback to improve usability and educational effectiveness. At the conclusion of development, a training workshop will be conducted for teachers covering system operation, student profile management, progress monitoring, and classroom integration. All student records, scores, and progress data will be stored locally with no cloud services or internet dependency at any stage.

2.2. Expected Deliverables

Please list the expected outcomes at the end of this project.

At the completion of the project, the following deliverables will be produced:

- **Fully functional offline desktop application:** A complete TypeCraft executable compatible with Windows 10 and Windows 11, optimised for smooth performance on low-end 4th-generation Intel hardware with integrated graphics.
- **Structured lesson system:** 5 progressive tiers (20+ lessons) beginning with home-row keys and advancing through top row, bottom row, speed building, and expert-level paragraphs. Lessons unlock sequentially based on accuracy performance.
- **Real-time typing engine:** Per-character green/red feedback, live WPM and accuracy display, combo streak tracking, mistake counting, and a countdown timer — all running at 30 FPS on low-end hardware.
- **Visual finger placement system:** An on-screen QWERTY keyboard with 8-colour finger coding that highlights the correct key on every keystroke and shows which finger the student should use.
- **Student profile system:** Individual profiles storing name, avatar, XP, level, lesson history, badges, daily streak, and typing statistics in a local SQLite database.
- **Gamification system:** XP gain with accuracy multipliers, 10 level tiers, 10 achievement badges, 3-star lesson ratings, daily streak tracking, and age-appropriate encouragement messages.
- **Classroom leaderboard:** Offline rankings by best WPM and best accuracy, covering all student profiles on the same computer with no internet required.
- **Teacher dashboard:** PIN-protected interface showing per-student average WPM, accuracy, lessons completed, and streak data. Includes the ability to reset individual student progress.
- **Windows executable (.exe):** PyInstaller-packaged single-folder distribution requiring no Python installation. Deployable via USB drive to any school computer.
- **Documentation:** Teacher quick-start guide, student usage guide, deployment instructions, and a lessons.json editing guide so teachers can add lesson content without coding.
- **Teacher training workshop:** An on-site session in the third or fourth week of August 2026 covering system setup, profile management, progress monitoring, leaderboard use, and basic troubleshooting.

2.3. Planned Schedule

Kindly list the start/end dates and the timeline for intermediate milestones.

Project Duration: 18th May 2026 — 18th August 2026

Dates	Phase	Activities
18 – 22 May 2026	Setup & Design	Requirement finalisation, system design, development environment setup, database schema design, lesson structure planning, and project scaffolding.
23 – 25 May 2026	Core Engine	Development of the typing engine: real-time input handling, WPM calculation, accuracy tracking, mistake detection, combo streak logic, and basic lesson loading system.
26 – 31 May 2026	Eid-ul-Adha Break	No development scheduled.

Dates	Phase	Activities
1 – 7 June 2026	UI Implementation	Main menu scene, student profile creation and selection, lesson selection interface, visual keyboard renderer with finger colour-coding, and results screen with star ratings and encouragement messages.
8 – 12 June 2026	Gamification & Data	XP and levelling system, badge award logic, daily streak tracking, classroom leaderboard, teacher dashboard with PIN protection, and full SQLite offline data storage integration.
13 – 15 June 2026	Testing & Optimisation	Full play-through testing of all 20 lessons, bug fixing, performance profiling on low-end hardware, dirty-rect rendering optimisation, font surface caching, and usability improvements.
16 – 18 June 2026	Packaging & Docs	PyInstaller .exe build, deployment folder preparation, teacher quick-start guide, student usage guide, and lessons.json editing documentation.
3rd – 4th Week August 2026	Deployment & Training	On-site visits 2–3 days per week (~80 min/session). Installation on school computers, teacher walkthrough session, supervised student test run, final refinements, and formal handover to The Bridge School.

2.4. Modus Operandi

Kindly explain the mode of operation — visit frequency, remote work arrangement, and supervision method.

The student has already visited the site to inspect the school's computer systems and to develop a clear understanding of the project's system requirements and intended application. Based on this initial assessment, the development phase will commence on 18th May 2026.

From 18th May through 18th June 2026, during the school vacation period, the student will work remotely. This phase covers the full design and development cycle: core engine implementation, UI development, gamification integration, SQLite data layer, performance optimisation, PyInstaller packaging, and documentation preparation. Regular progress updates will be shared with supervisors for feedback and guidance.

In the third and fourth weeks of August 2026, the student will visit the school site 2 to 3 days per week, with each session lasting approximately 80 minutes. During these visits, the student will install the application on school computers, conduct a formal teacher walkthrough covering system usage and troubleshooting, run a supervised live session with students, and address any final refinements identified in the real classroom environment. The student will be overseen by the Team Leader through scheduled meetings, attendance monitoring, and performance evaluations.

2.5. Comments

Kindly mention any special requirements or additional comments.

Both supervisors reviewed the features, functionality, and overall scope of the project in detail and had no additional requirements or comments at the time of submission.

Name & Signature of Student

SHANFEER QURESHI

Rayyan Amir Alvi

Muhtees

Location: Karachi

Date: 14/5/2026

Name & Signature of External Supervisor

Heera Abid

Fanat

1) Miss Hira Hunzai

2) Miss Farah Mehdavi

(The Bridge School)

Location: Karachi

Date: May 14, 2026

Name & Signature of Internal Supervisor

Farah Mehdavi

Location: _____

Date: _____